

Regression Analysis of the MathExam14W Data

Two Bern-Bino regression variants versus logistic and beta-binomial regression

This report documents the regression analysis of the MathExam14W data set. Two variants of the Bern-Bino regression of Section 2.5 are compared with logistic regression and beta-binomial regression: a fully component-specific variant, in which every Bernstein component has its own regression vector, and the parsimonious variant of Section 2.5, in which the components share one regression vector and differ through a component-specific intercept and scale. The data are split once into a training set and a held-out test set, all models are estimated on the training set only, and the Bernstein degree of each variant is selected inside the training set by cross-validation. Section 1 introduces the data and the design of the experiment, Section 2 defines the variables, Section 3 states the models and the selection of the Bernstein degree, Section 4 reports the estimation results on the training set, Section 5 reports the validation on the held-out test set and Section 6 concludes.

1 Data and design of the experiment

1.1 The data set

The data are the individual results of the course ‘Mathematics 101’ for first-year business and economics students at the University of Innsbruck, Austria (Zeileis, Umlauf and Leisch, 2014). The course assessment combined biweekly online exercises, consisting of 26 numerical exercises administered through the OpenOLAT platform, with a written final examination of 13 single-choice items, each offering five answer alternatives; an incorrect answer was penalised by 25 per cent of the item value in order to discourage random guessing. Of the 941 students registered for the course, $n = 729$ took the examination. Because of the large number of participants the students were divided into two groups that received partially different versions of items 1, 5, 6, 7, 8, 9, 11 and 12; the versions differed in the story or in the numerical set-up but not in the mathematical skills required to solve them.

Treating the 13 items as Bernoulli trials, the response of student i is X_i , the number of items solved correctly, so that X_i takes values in $\{0, 1, \dots, 13\}$ and $X_i | p_i \sim \text{Binomial}(13, p_i)$, with the same number of trials $m = 13$ for every student and p_i the student-specific ability. The score distribution is unimodal and slightly right-skewed, with the mode at $X = 7$ and $X = 8$; 9 students solved no item at all and 32 students solved all 13 items. The proportions of correct answers differ considerably across items, ranging from 17.4 to 78.5 per cent, which indicates substantial heterogeneity in ability.

1.2 Training and test sets

The 729 observations are split *once* at random into a training set of $n_1 = 511$ students (70.1 per cent) and a held-out test set of $n_2 = 218$ students (29.9 per cent), with a fixed random seed. The test set is separated from the very beginning and takes no part in the estimation of any model nor in the selection of the Bernstein degree; it is used only for the final evaluation. Table 1 reports the score distribution of the two subsets, which are closely comparable.

Table 1. Score distribution in the training and test sets.

X	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Training, n	8	13	18	25	34	38	57	64	79	56	42	36	20	21
Training, %	1.6	2.5	3.5	4.9	6.7	7.4	11.2	12.5	15.5	11.0	8.2	7.0	3.9	4.1
Test, n	1	3	6	8	16	27	14	40	19	25	25	14	9	11

X	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Test, %	0.5	1.4	2.8	3.7	7.3	12.4	6.4	18.3	8.7	11.5	11.5	6.4	4.1	5.0

2 Variables

The response is *nsolved*, the number of items solved correctly. Four covariates are used: *tests*, the number of online test exercises solved correctly before the examination, which measures study effort; *gender*; *study*, the type of degree programme; and *semester*, the number of semesters enrolled in the programme. Table 2 summarises them.

Table 2. Response and covariates used in the regression analysis.

Variable	Type	Description	Summary (n = 729)
nsolved	count	Response: number of items solved correctly (0-13)	mean 7.52, mode 7
tests	continuous	Covariate: online exercises solved correctly before the examination	range 9-26, mean 19.63
gender	binary	Covariate: student gender	326 female, 403 male
study	binary	Covariate: 571 = three-year bachelor, 155 = four-year diploma	583 bachelor, 146 diploma
semester	count	Covariate: number of semesters enrolled	range 1-21, median 1

Let z_i denote the four covariates of student i , with the two continuous covariates standardised. For the two Bern-Bino variants we write ζ_i for the covariate vector including an intercept and z_i for the covariates alone (no intercept). The distinction matters: in the parsimonious variant the level of each component is carried by its own intercept φ_k , so an intercept must **not** be included in the shared regression vector. The data set also contains *attempt* and *group*; neither is used, because the five levels of *attempt* are highly unbalanced (431 of the 729 students sat the examination for the first time) and *group* describes the administration of the examination rather than a characteristic of the student.

3 Models and selection of the Bernstein degree

3.1 The four regression models

All four models describe binomial counts with a fixed number of trials, $X_i | p_i \sim \text{Binomial}(13, p_i)$, and differ in the way the student-specific probability p_i is specified.

(i) Logistic regression. $\text{logit}(p_i) = \zeta_i^T \alpha$: the success probability is common to all items of a student and varies across students only through the covariates.

(ii) Beta-binomial regression. $p_i \sim \text{Beta}(a_i, b_i)$ with $\text{logit}(\mu_i) = \zeta_i^T \alpha$, $\mu_i = a_i / (a_i + b_i)$, $a_i = \mu_i(1 - \rho)/\rho$ and $b_i = (1 - \mu_i)(1 - \rho)/\rho$, with a constant dispersion parameter ρ .

(iii) Bern-Bino regression, variant A (component-specific regression). $p_i \sim \text{Bernstein}(K, \lambda_i)$ with

$$\lambda_{ik} = \exp(\eta_{ik}) / \sum_l \exp(\eta_{il}), \quad \eta_{ik} = \zeta_i^T \beta_k, \quad k = 0, \dots, K,$$

that is every Bernstein component has its own regression vector β_k . Because adding the same vector to all β_k leaves every λ_{ik} unchanged, one component must serve as a reference; we set $\beta_0 = 0$, which leaves Kp free parameters with $p = 5$.

(iv) Bern-Bino regression, variant B (Section 2.5; shared regression with component-specific scales). $p_i \sim \text{Bernstein}(K, \lambda_i)$ with

$$\lambda_{ik} = \exp(\eta_{ik}) / \sum_l \exp(\eta_{il}), \quad \eta_{ik} = \varphi_k + \gamma_k (\zeta_i^T \beta), \quad k = 0, \dots, K,$$

with $\varphi_0 = 0$ and $\gamma_0 = 1$. Here all components **share one regression vector** β , which acts on the covariates only (the intercept is absorbed by φ_k), and each component applies its own scale γ_k to the shared index. The number of free parameters is $2K + 4$. Variant B therefore replaces the K separate vectors of variant A by one shared vector plus K scales, which is the parsimonious specification proposed in the thesis. In both variants the prior density of the ability is a Bernstein polynomial whose shape, and not only its mean, depends on the covariates.

3.2 Selection of the Bernstein degree K

For **both variants** the degree K is selected **inside the training set** only, over the grid $K = 1, \dots, 10$, by **five-fold cross-validation repeated twice**: the 511 training observations are divided into five folds, the model is estimated on four fifths of them and evaluated on the remaining fifth, and the ten resulting values (five folds $\times$ two repetitions) are averaged for each candidate K . **The criterion used for the selection is the validation log-likelihood**; for completeness the validation mean squared error (MSE) and the training-set BIC are also recorded. Figure 1 shows the three criteria for the two variants.

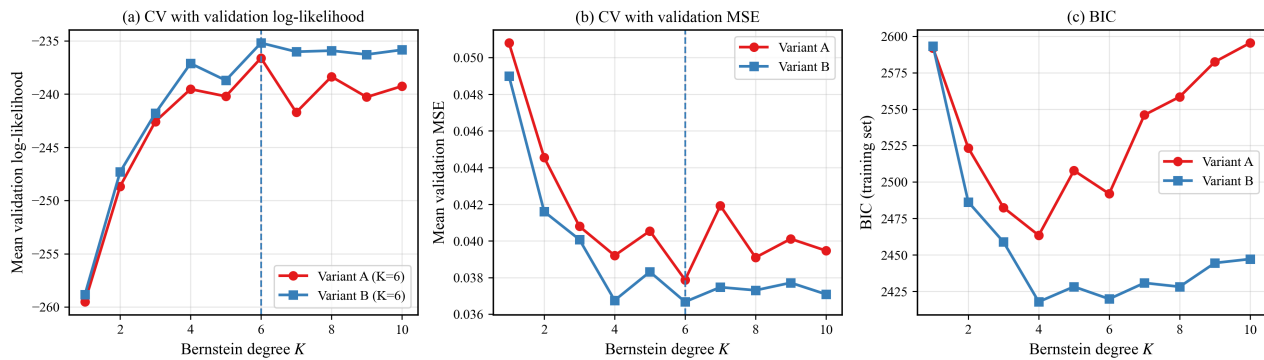


Figure 1. Selection of the Bernstein degree K on the training set for the two variants: (a) five-fold repeated cross-validation with the validation log-likelihood; (b) the same cross-validation with the validation MSE; (c) the training-set BIC.

Both variants select $K = 6$ by the cross-validation log-likelihood, and the cross-validation MSE selects the same degree for both; the BIC, which penalises the additional weight parameters more heavily, selects $K = 4$ for both. The agreement of the two cross-validation criteria is reassuring, and the models reported below use $K = 6$, giving 30 parameters for variant A (6×5) and 16 for variant B ($2 \times 6 + 4$).

4 Estimation results on the training set

All models are estimated by maximum likelihood on the 511 training observations. The standard errors of the logistic regression are those of the fitted generalised linear model and those of the beta-binomial regression come from the numerically differentiated Hessian of its log-likelihood. The likelihood of a Bern-Bino regression is highly multimodal in the parameters, so a multi-start strategy is used for both variants: the optimizer is started repeatedly from deterministic and random starting values and the solution with the largest likelihood is retained.

4.1 Logistic and beta-binomial regression

Table 3. Logistic regression (training set, $n = 511$).

Term	Estimate	S.E.	z	p -value
Intercept	0.3675	0.0635	5.79	$< 1e-8$
<i>tests</i>	0.5655	0.0272	20.82	$< 1e-96$
<i>genderM</i>	-0.0154	0.0541	-0.29	0.775
<i>study571</i>	-0.1121	0.0691	-1.62	0.105

Term	Estimate	S.E.	z	p -value
<i>semester</i>	0.1459	0.0270	5.41	< 1e-7

Table 4. Beta-binomial regression (training set, $n = 511$).

Term	Estimate	S.E.
Intercept	0.3729	0.0912
<i>tests</i>	0.5748	0.0391
<i>genderM</i>	-0.0278	0.0784
<i>study571</i>	-0.1104	0.0997
<i>semester</i>	0.1403	0.0390
logit(ρ)	-2.2884	0.1156

The number of online exercises solved before the examination has by far the strongest effect in both models; gender and the type of degree programme are not significant at the 5 per cent level. The dispersion parameter of the beta-binomial regression is $\rho = \text{expit}(-2.2884) = \mathbf{0.0921}$, clearly positive, which confirms the presence of extra-binomial variation: the ability varies more between students than a binomial model with the same mean allows. Accounting for this variation raises the training log-likelihood from -1275.13 to -1173.31.

4.2 Bern-Bino regression, variant A ($K = 6$)

Variant A estimates a separate regression vector for each of the six non-reference components, so that it uses 30 parameters. Table 5 reports the estimates.

Table 5. Variant A ($K = 6$, 30 parameters): component-specific regression coefficients β_k ; $\beta_0 = 0$ is the reference component.

Term	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
Intercept	-3.75	-194.84	289.63	305.15	-122.45	-207.68
<i>tests</i>	-11.54	-574.23	-199.22	39.36	363.52	402.62
<i>genderM</i>	-2.89	98.93	46.19	-52.53	-4.93	-46.23
<i>study571</i>	-2.29	57.52	11.01	21.55	-52.99	20.39
<i>semester</i>	-4.24	-132.13	-65.33	52.16	95.17	64.70

4.3 Bern-Bino regression, variant B ($K = 6$)

Variant B replaces the six separate vectors by one shared vector, one intercept and one scale per component, giving $2K + 4 = 16$ parameters. Tables 6 and 7 report the estimates; $\varphi_0 = 0$ and $\gamma_0 = 1$ are the identifying constraints and are not estimated.

Table 6. Variant B ($K = 6$): component intercepts and scales.

k	0	1	2	3	4	5	6
φ_k	0 (fixed)	4.864	-298.780	347.542	205.843	-123.152	-131.355
γ_k	1 (fixed)	-15.597	-19.754	-10.764	19.665	23.914	24.018

Table 7. Variant B: shared regression coefficients β (the intercept is not part of β).

Term	Estimate	Term	Estimate
<i>tests</i>	56.260	<i>study571</i>	2.048
<i>genderM</i>	-0.757	<i>semester</i>	4.921

Remarks on the estimates. Two features are worth recording. First, the coefficients of both Bern-Bino variants are numerically large. This is a property of the fitted mixture rather than an error: the components take extreme values of the linear index so that the softmax weights become a sharp function of the covariates, which is what allows the shape of the prior to change substantially across students. Only the products $\gamma_k \beta$ in variant B, and the differences $\beta_k - \beta_0$ in variant A, are directly interpretable; both indicate that the prior of a component varies strongly with study effort. Second, the specification of variant B must keep the intercept out of the shared vector β : with φ_k carrying the level of component k , an intercept inside β would be confounded with φ_k and would introduce a redundant direction. With the correct specification the constraint $\gamma_0 = 1$ identifies the scale of β , because the softmax weights are strictly positive and hence the first component always carries weight.

Finally, several Bernstein components receive a numerically negligible weight in the fitted models. The parameters of a component that is never used are not identified, so the Hessian of the log-likelihood is singular and standard errors are not available for the Bern-Bino coefficients of either variant. This is a well-known feature of over-specified finite mixtures and it does not affect the fitted predictive distribution, which is the object of interest here.

5 Validation on the held-out test set

The four fitted models are applied to the 218 students of the test set, which took no part in the estimation. The models are compared by the log-likelihood evaluated on the test set, and by the mean absolute error (MAE) and the mean squared error (MSE) between the observed proportion of solved items $x_i/13$ and the proportion predicted by the model. The information criteria are computed from the test log-likelihood as $AIC = 2k - 2\ell$ and $BIC = k \log n_2 - 2\ell$, where k is the number of parameters estimated on the training set and ℓ the test log-likelihood; they combine the out-of-sample fit with a penalty for complexity.

Table 8. Fit on the training set and validation on the held-out test set ($n_1 = 511, n_2 = 218$).

Model	k	K	tr. logLik	tr. AIC	tr. BIC	te. logLik	te. AIC	te. BIC	te. MSE	te. MAE
Logistic	5	-	-1275.13	2560.26	2581.45	-575.23	1160.46	1177.38	0.04144	0.1668
Beta-binomial	6	-	-1173.31	2358.63	2384.04	-516.40	1044.81	1065.11	0.04156	0.1669
Bern-Bino A	30	6	-1150.61	2361.21	2488.31	-509.64	1079.28	1180.81	0.04027	0.1644
Bern-Bino B	16	6	-1153.22	2338.43	2406.21	-507.32	1046.65	1100.80	0.03917	0.1621

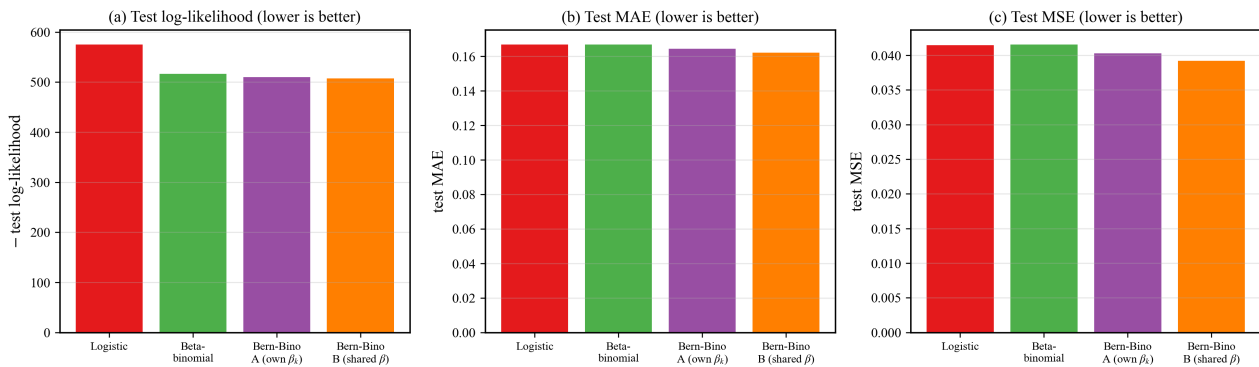


Figure 2. Held-out test-set performance of the four models: (a) test log-likelihood, (b) test MAE and (c) test MSE.

The two Bern-Bino variants are the only models that improve substantially on the beta-binomial regression out of sample. **Variant B, the parsimonious specification of Section 2.5, performs best on all three test criteria:** its test log-likelihood is -507.32 against -509.64 for variant A, -516.40 for the beta-binomial and -575.23 for the logistic regression, and it also attains the smallest test MSE (0.03917) and MAE (0.1621). Relative to variant A it gains 2.32 log-likelihood units on the test set while using 16 instead of 30 parameters, so the shared vector β with component-specific scales γ_k captures the covariate-dependent shape of the prior at least as well as a full set of component-specific regressions, and considerably more efficiently. The gain of variant B over the best parametric competitor, 9.08 log-likelihood units, is obtained on data that were never used in the estimation and therefore reflects a genuine improvement in out-of-sample fit.

The information criteria that penalise complexity tell a partially different story. The beta-binomial regression still has the smallest test BIC (1065.11 against 1100.80 for variant B), whereas the test AIC of variant B (1046.65) is essentially the same as that of the beta-binomial regression (1044.81) despite the ten additional parameters. Variant B is therefore the best model when the accuracy of the fitted distribution matters, whereas the beta-binomial model remains attractive if parsimony is the primary concern.

6 Conclusions

This analysis compares two Bern-Bino regression variants with logistic regression and beta-binomial regression on the MathExam14W data, which consist of the number of correctly solved items of 729 students in a 13-item examination and thus of binomial counts with a fixed number of trials, $m = 13$. Variant A gives every Bernstein component its own regression vector β_k , whereas variant B, the specification proposed in Section 2.5, uses one shared regression vector β applied to the covariates together with a component-specific intercept φ_k and scale γ_k . The models were estimated on a training set of 511 students, the degree of each variant was selected inside the training set by five-fold cross-validation repeated twice using the validation log-likelihood, and all models were evaluated on a held-out test set of 218 students.

The main findings are the following.

- (i) **Extra-binomial variation is present and important.** The dispersion parameter of the beta-binomial regression is clearly positive ($\rho = 0.0921$), and accounting for it improves the training log-likelihood from -1275.13 (logistic) to -1173.31.
- (ii) **Both Bern-Bino variants give the best out-of-sample fit.** On the test set they attain the two largest log-likelihoods (-507.32 for variant B and -509.64 for variant A) and the two smallest values of the test MSE and MAE, clearly improving on the logistic and beta-binomial regressions. The improvement is therefore not an artefact of in-sample overfitting.
- (iii) **The parsimonious variant B dominates the fully component-specific variant A.** With 16 instead of 30 parameters, variant B achieves a better test log-likelihood by 2.32 units, a smaller test MSE (0.03917 against 0.04027) and a smaller test MAE (0.1621 against 0.1644), and it also has the better training AIC and BIC. Sharing one regression vector while letting each component choose its own intercept and scale is thus not merely a device for saving parameters: it regularises the covariate effect and improves prediction.
- (iv) **The improvement of variant B is paid for by additional parameters.** The test BIC still favours the beta-binomial regression (1065.11 against 1100.80), while the test AIC of the two models is nearly identical (1046.65 against 1044.81) even though variant B uses ten more parameters. The Bernstein model is preferable when the accuracy of the fitted distribution matters, whereas the beta-binomial model remains attractive when parsimony is the primary concern.
- (v) **The specification of the shared vector matters.** In variant B the intercept must be carried by φ_k and kept out of β ; otherwise the two are confounded and a redundant direction appears. With the correct specification the constraint $\gamma_0 = 1$ identifies the scale of β , since the softmax weights are strictly positive. Standard errors for the Bern-Bino coefficients are nevertheless unavailable, because components that receive zero weight leave their own parameters unidentified and the Hessian is singular; the fitted predictive distribution and all test-set comparisons above are unaffected.

References

Zeileis, A., Umlauf, N. and Leisch, F. (2014). Flexible generation of e-learning exams in R: Moodle quizzes, OLAT assessments, and beyond. *Journal of Statistical Software*, 58(1), 1-36.